

MDTS4214_711_PS_7_full

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Problem Set 7: Shrinkage Methods

We would be working with the Hitters dataset available in the ISLR library. We have to predict the Salary based on all available variables. First, we make sure that we delete all rows with one or more missing values.

1. We split the data set into a training set and a test set randomly, keeping 100 observations in the test set and the remaining in the training set.

```
set.seed(123)
rm(list=ls())
library(glmnet)
library(ISLR)
```

```
Hitters = na.omit(Hitters)
```

```
test_index = sample(1:nrow(Hitters), 100)
```

```
test_data = Hitters[test_index, ]
train_data = Hitters[-test_index, ]
```

2. We train the model using least squares and report the train MSE and test MSE.

```
lm_model = lm(Salary ~ ., data = train_data)
```

```
y_hat = predict(lm_model, newdata = train_data)
y_test = predict(lm_model, newdata = test_data)
```

```
train_mse = mean((train_data$Salary-y_hat)^2)
train_mse
```

```
## [1] 105334.2
```

```
test_mse = mean((test_data$Salary-y_test)^2)
test_mse
```

```
## [1] 87803.91
```

3. We fit ridge regression models on the training set, with grid values of λ . We report the following:

- (i) the dimension of the coefficient matrix so obtained.
- (ii) The l_2 norm of the regression coefficients corresponding to the 50th and 60th choice of λ . We comment on the output.
- (iii) Test MSE corresponding to these two choices of λ .

```

lambda_grid = 10^seq(5,-2, length = 100)
x_train = model.matrix(Salary~.,train_data)[,-1]
x_test = model.matrix(Salary~.,test_data)[,-1]
head(x_train)

##              AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun
## -Alan Ashby      315   81     7  24  38   39   14   3449   835    69
## -Alvin Davis     479  130    18  66  72   76    3   1624   457    63
## -Andre Dawson    496  141    20  65  78   37   11   5628  1575   225
## -Alfredo Griffin 594  169     4  74  51   35   11   4408  1133    19
## -Andres Thomas   323   81     6  26  32    8    2    341    86     6
## -Andre Thornton  401   92    17  49  66   65   13   5206  1332   253
##              CRuns CRBI CWalks LeagueN DivisionW PutOuts Assists Errors
## -Alan Ashby      321  414   375      1      1      632    43    10
## -Alvin Davis     224  266   263      0      1     880    82    14
## -Andre Dawson    828  838   354      1      0     200    11     3
## -Alfredo Griffin 501  336   194      0      1     282   421    25
## -Andres Thomas    32   34     8      1      1     143   290    19
## -Andre Thornton  784  890   866      0      0       0     0     0
##              NewLeagueN
## -Alan Ashby           1
## -Alvin Davis          0
## -Andre Dawson         1
## -Alfredo Griffin      0
## -Andres Thomas        1
## -Andre Thornton       0

head(x_test)

##              AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun CRuns
## -Marvell Wynne    288   76     7  34  37   15    4   1644   408    16   198
## -Bill Almon       196   43     7  29  27   30   13   3231   825    36   376
## -Rick Cerone      216   56     4  22  18   15   12   2796   665    43   266
## -Jack Howell      151   41     4  26  21   19    2    288    68     9    45
## -Spike Owen       528  122     1  67  45   51    4   1716   403    12   211
## -Tom Paciorek     213   61     4  17  22    3   17   4061  1145    83   488
##              CRBI CWalks LeagueN DivisionW PutOuts Assists Errors NewLeagueN
## -Marvell Wynne    120   113      1      1     203     3     3      1
## -Bill Almon       290   238      1      0     80     45    8      1
## -Rick Cerone      304   198      0      0    391     44    4      0
## -Jack Howell       39    35      0      1     28     56    2      0
## -Spike Owen       146   155      0      1    209    372   17      0
## -Tom Paciorek     491   244      0      1    178     45    4      0

ridge_model=glmnet(x_train, train_data$Salary, alpha = 0,lambda= lambda_grid)
dim(coef(ridge_model))

## [1]  20 100

```

(i) Dimension of Coefficient Matrix

```

dim(coef(ridge_model))

## [1]  20 100

```

(ii) Coefficients for 50th and 60th value of λ respectively

```
cat("50th value\n\n")

## 50th value

coef_50 = coef(ridge_model)[-1, 50]
coef_50

##           AtBat           Hits           HmRun           Runs           RBI
## -0.56861594    1.57781724    1.45165595    0.26148608    2.54343128
##           Walks           Years           CAtBat           CHits           CHmRun
##  2.88812942   -4.15498762    0.01212119    0.12212743    0.02430066
##           CRuns           CRBI           CWalks           LeagueN           DivisionW
##  0.22110562    0.24061114   -0.27212940   54.66429035  -145.12121490
##           PutOuts           Assists           Errors           NewLeagueN
##  0.35444338    0.26953832   -5.05469505  -14.05949208

cat("\n\n60th value\n\n")

##
##
## 60th value

coef_60 = coef(ridge_model)[-1, 60]
coef_60

##           AtBat           Hits           HmRun           Runs           RBI
## -1.408336e+00  3.329840e+00  1.626042e+00 -1.917524e-01  3.107071e+00
##           Walks           Years           CAtBat           CHits           CHmRun
##  4.844059e+00 -7.920993e+00 -5.315477e-03  1.602586e-01 -3.013027e-01
##           CRuns           CRBI           CWalks           LeagueN           DivisionW
##  4.667053e-01  3.938615e-01 -6.029094e-01  5.686263e+01 -1.526506e+02
##           PutOuts           Assists           Errors           NewLeagueN
##  3.918256e-01  3.813593e-01 -5.711395e+00 -2.698770e+01
```

(iii). L2 Norm of Coefficients

```
l2_norm = function(beta){
  sqrt(sum(beta^2))
}

# L2 norm of 50th Lambda
l2_50 = l2_norm(coef_50)
l2_50

## [1] 155.9135

# L2 norm of 60th Lambda
l2_60 = l2_norm(coef_60)
l2_60

## [1] 165.5575
```

(iv). Test MSE for 50th and 60th λ

```
# Predictions
ridge_pred_50 = predict(ridge_model, s = lambda_grid[50], newx = x_test)
ridge_pred_60 = predict(ridge_model, s = lambda_grid[60], newx = x_test)

# Test MSE for 50th Lambda
Test_mse_50 = mean((test_data$Salary - ridge_pred_50)^2)
Test_mse_50

## [1] 89317.28

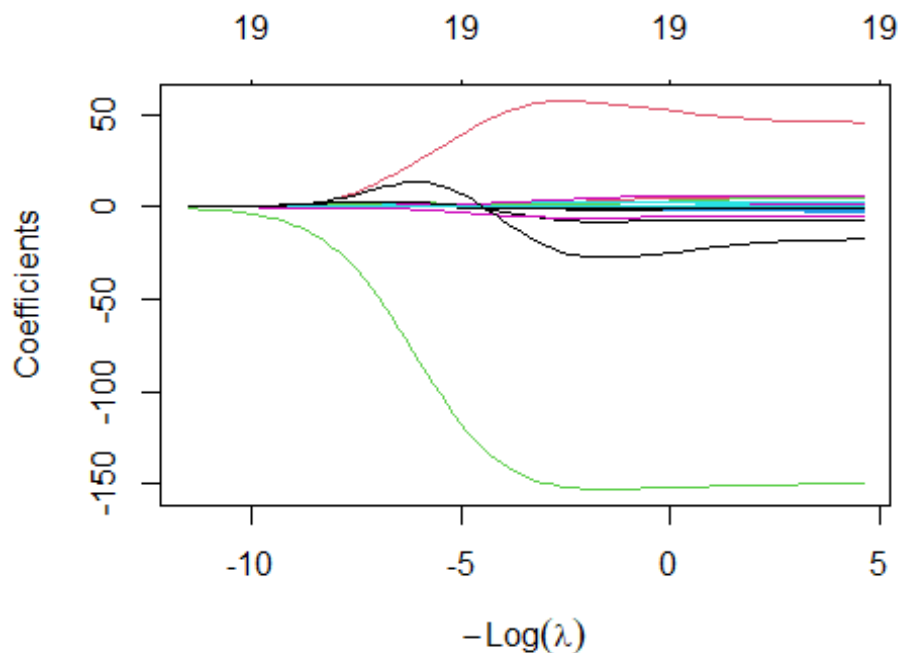
# Test MSE for the 60th Lambda
Test_mse_60 = mean((test_data$Salary - ridge_pred_60)^2)
Test_mse_60

## [1] 87580.18
```

We observe that the test MSE is dropping as λ increases.

4. Make a line plot (for all the predictors) of the standardized coefficients against the grid value of λ chosen by you. Interpret the graph.

```
plot(ridge_model)
```



Interpretation: Here we obtain the values of $-\log(\lambda)$ on X-axis. Each of the line plots denote the respective coefficients with respect to $-\log(\lambda)$.

- We know that the increasing value of λ the value of $-\log(\lambda)$ decreases.
- Hence we can say for the increasing value of λ , the coefficients are shrinking.

5. How would the graph in (4) change if instead of λ you plot on the x axis

the ratio of l2 norm of the ridge regression coefficient to the least square regression coefficient

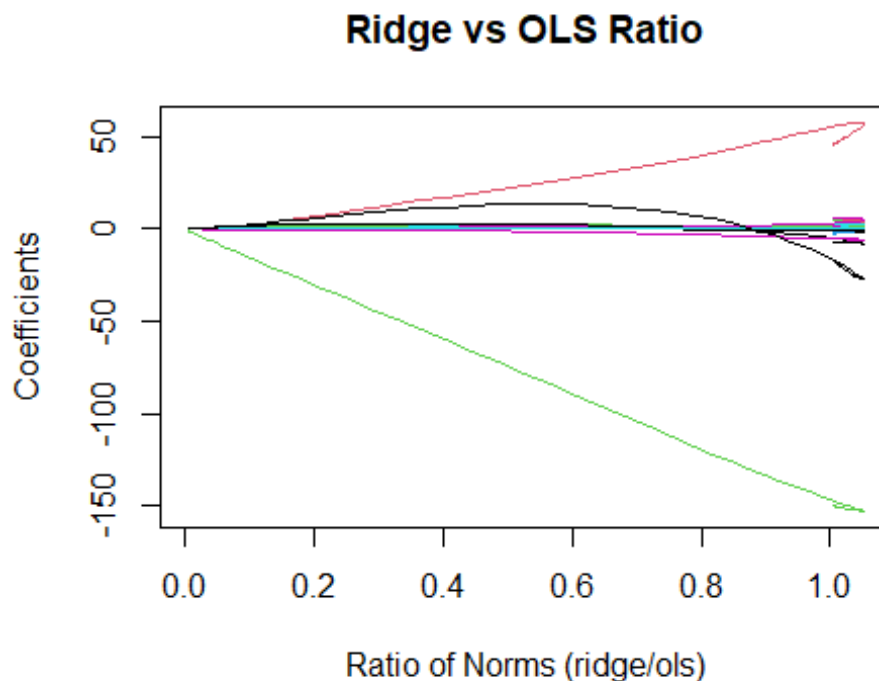
```
beta_ols = coef(lm_model)[-1]

ridge_coefs = as.matrix(coef(ridge_model))[-1, ]

ridge_norms = apply(ridge_coefs, 2, l2_norm)
ols_norm     = l2_norm(beta_ols)

ratio = ridge_norms / ols_norm

matplot(ratio, t(ridge_coefs), type = "l", lty = 1,
        xlab = "Ratio of Norms (ridge/ols)", ylab = "Coefficients",
        main = "Ridge vs OLS Ratio")
```



6. Run the Ridge regression on training data by choosing λ by cross-validation. Find the test MSE for the optimal choice of λ obtained using cross validation.

```
set.seed(123)
cv_ridge = cv.glmnet(x_train, train_data$Salary, alpha = 0)
cv_ridge

##
## Call:  cv.glmnet(x = x_train, y = train_data$Salary, alpha = 0)
##
## Measure: Mean-Squared Error
##
##      Lambda Index Measure      SE Nonzero
## min      404      69 141709 32690      19
## 1se     7937      37 173528 39657      19

lambda_ridge = cv_ridge$lambda.min
cat("\n\nOptimum Lambda for Ridge Model: ",lambda_ridge,"\n\n")

##
##
## Optimum Lambda for Ridge Model:  404.3208

ridge_cv_pred = predict(cv_ridge,"lambda.min", newx = x_test)
ridge_coef = coef(cv_ridge,"lambda.min")

mse_ridge_cv = mean((test_data$Salary - ridge_cv_pred)^2)
non_zero = sum(ifelse(ridge_coef[-1] != 0,1,0))

list(Ridge_Test_MSE = mse_ridge_cv, Nonzero_Coefficients = non_zero)

## $Ridge_Test_MSE
## [1] 93690.09
##
## $Nonzero_Coefficients
## [1] 19
```

7. Fit a LASSO model on the training set, with λ chosen by cross-validation. Report the test MSE, along with the number of non-zero coefficient estimates.

```
set.seed(123)
cv_lasso = cv.glmnet(x_train, train_data$Salary, alpha = 1)
cv_lasso

##
## Call:  cv.glmnet(x = x_train, y = train_data$Salary, alpha = 1)
##
## Measure: Mean-Squared Error
##
##      Lambda Index Measure      SE Nonzero
```

```
## min 16.71    29 150268 32062      8
## 1se 141.96    6 179980 41276      2

lambda_lasso = cv_lasso$lambda.min
cat("\n\nOptimum Lambda for Lasso Model: ", lambda_lasso, "\n\n")

##
##
## Optimum Lambda for Lasso Model: 16.70659

lasso_cv_pred = predict(cv_lasso, "lambda.min", newx = x_test)
lasso_coef = coef(cv_lasso, "lambda.min")

mse_lasso_cv = mean((test_data$Salary - lasso_cv_pred)^2)
non_zero = sum(ifelse(lasso_coef[-1] != 0, 1, 0))

list(Lasso_Test_MSE = mse_lasso_cv, Nonzero_Coefficients = non_zero)

## $Lasso_Test_MSE
## [1] 93914.89
##
## $Nonzero_Coefficients
## [1] 8
```

8. Fit an Elastic Net model on the training set, with λ_1 and λ_2 chosen by cross-validation. Report the test MSE. Also report the predictors with non-zero coefficients.

```
set.seed(123)
alpha = seq(0, 1, by = 0.1)

cv_errors = c()
models = vector("list")

for(i in 1:length(alpha))
{
  models[[i]] = cv.glmnet(x_train, train_data$Salary, alpha = alpha[i])
  cv_errors[i] = min(models[[i]]$cvm)
}

best_id = which.min(cv_errors)
best_alpha = alpha[best_id]
best_model = models[[best_id]]

lambda_elastic = best_model$lambda.min
cat("\n\nOptimum Lambda for Elastic Net Model: ", lambda_elastic, "\n\n")

##
##
## Optimum Lambda for Elastic Net Model: 95.60123
```

```

pred_elastic = predict(best_model,"lambda.min", newx = x_test)
mse_elastic  = mean((test_data$Salary - pred_elastic)^2)

coef_elastic = coef(best_model,"lambda.min")
nz_vars = rownames(coef_elastic)[coef_elastic[,1] != 0][-1]

list(Elastic_Test_MSE = mse_elastic, Non_Zero_Predictors = nz_vars)

## $Elastic_Test_MSE
## [1] 91957.09
##
## $Non_Zero_Predictors
## [1] "Hits"      "HmRun"     "RBI"       "Walks"     "CAtBat"    "CHits"
## [7] "CHmRun"    "CRuns"     "CRBI"      "LeagueN"   "DivisionW" "PutOuts"
## [13] "Assists"   "Errors"

```

9. Compare the test MSE 's of least squares regression, Ridge regression, LASSO and ELASTIC NET and comment on how accurately we can predict the Salary of a baseball player.

Test MSE values:

- Least Squares: 87803.91
- Ridge: 93690.09
- LASSO: 93914.89
- Elastic Net: 91957.09

We observe that: - The test MSE is least for the OLS method. - Out of the regularized methods, Elastic Net performs the best. - LASSO and Elastic Net generate simpler models by reducing the number of predictor, which makes the models more interpretable. - Overall for pure accuracy, OLS should be preferred while regularization can be used for dimensionality reduction and superior interpretability.
